Gravitational field strength

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a gravitational field to predict the force on a test mass.

SUCCESS CRITERIA - I can...

1. Represent field direction with arrows.
2. Use g=F/m and F=mg consistently.
3. Distinguish source field from the mass placed in it.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “field” and “gravity” to explain your model or result. Sentence starters are optional.

field: a description of an influence at each location

gravity: an attractive interaction between masses

radius: distance from the center to the edge of a circle or sphere

01 Bring back an earlier idea

An object weighs 29.4 N where g=9.8 N/kg.
Find its mass and its weight in a field of 5 N/kg.

Word help: weight = the gravitational force on an object

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $g = F_g/m = GM/r^2$.
Outside a spherical source of mass M; r is distance from its center. Field direction is toward the source.

Worked reasoning: A 2 kg test object feels 6 N toward a source. The local field is g=6/2=3 N/kg. A 4 kg test object at that point feels 12 N.

Your turn: Draw the field arrow and the two possible force arrows. Which quantity belongs to the location rather than the test mass?

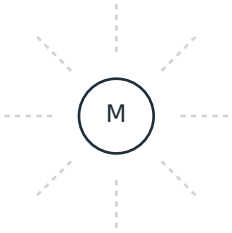
Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

Draw field arrows at three locations around the source. Which way would a small test mass accelerate?

Sentence starter: The ... in my model represents ... . Its label shows ... .



Add arrowheads to show field direction.

The center marks the source; add arrowheads on the surrounding lines to show field direction.

Marzano check - circle: ① ② ③

04 Solve and justify

A test mass .20 kg experiences .60 N.
Find g. If source GM=12 m^3/s^2, find distance from the source center at that field.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

Gravitational field strength

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

A student says a heavier test mass falls faster because it feels more gravitational force.
Use F=mg and a=F/m to respond.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Choose a method

A 4 kg block slides down a 30° ramp with 11.6 N friction up the ramp.
Use g=9.8 m/s².
Find along-ramp acceleration and normal force.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

07 Exit check

Outside a spherical source, field is 8 N/kg at r.
Find field at 2 r and force on .50 kg there.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

An exterior field is 8 N/kg at distance r from a spherical source. Where is it 2 N/kg? What force acts there on a 0.25 kg mass?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3